

Lecture Notes on QFT I

October 23, 2025

Readings

- Peskin & Schroeder Sec. 2.1 and 2.2
- Weinberg vol 1 Chap. 1

Lecture 1: Principle of Locality and Fields

The goals of this Quantum Field Theory class is

- To develop concepts, formalism and techniques for quantum dynamics of fields.
- Quantum Field Theory is also very important for other interactions in nature. Among the four fundamental interactions in nature, three of them are described by QFT completely.
- Even though QFT is initially developed for particle physics, but over the years, QFT has been found many applications in many other branches of physics, in condensed matter, statistical physics and so on. It's fair to say, nowadays, QFT has become a universal language for theoretical physics in all different fields.

Why we consider QFT?

principle of locality

The first important concept is called the **principle of locality**.

In Newtonian mechanics, you have action at a distance, for example, if you look at gravity and the gravity is exerted by the sun on the Earth, but they are very far away. And the same thing with the Coulomb interaction between the charged particles.

But then, in the 19th century, they came from this principle of locality, that is formulated by Faraday around 1830. The principle of locality says that all points you don't have action at a distance. He said **all points in space participate in the physical process, and the effect propagates from points to a neighboring points**. So in this principle of locality, you don't have action at a distance, action at a distance is always conveyed from one point to another point through the propagating in the space.

The concepts of **fields** is **the mathematical device or vehicle that the principle locality is at work**. The main idea of the field is that we associate each point in space a dynamical variable.

For example, if you have electric field is defined at each point x , we can introduce an electric field, which can also depend on time. Of course, normally we write it this way $\vec{E}_{\vec{x}}(t) \equiv \vec{E}(\vec{x}, t)$. The reason I write it in this way $\vec{E}_{\vec{x}}(t)$ to emphasize that in the definition of the electric field, the space and times actually play very different roles. The space plays the role of a label. So at each point $\vec{x}$, we have electric field, so x here is just a label, and t is used to describe the evolution, the change of the electric field. So $\vec{x}$ and t play very different roles.

Similarly, you can do this to magnetic field $\vec{B}_{\vec{x}}(t) \equiv \vec{B}(\vec{x}, t)$, you should always treat $\vec{x}$ as labels. And we know the evolution of electric and magnetic field is described by Maxwell's equations,

$$\begin{cases} \nabla \cdot \vec{B} = 0 \\ \nabla \cdot \vec{E} = 4\pi\rho \\ \frac{\partial \vec{B}}{\partial t} = -\nabla \times \vec{E} \\ \nabla \times \vec{B} = 4\pi\vec{J} + \frac{\partial \vec{E}}{\partial t} \end{cases} \quad (1)$$

which is called the differential form of the Maxwell's equations. This set of equations exemplifies perfectly the principle of locality. It's because you see those equations only involve the value of electric fields and the magnetic field at a single point. If $\vec{B}$ is at $\vec{x}$ point, then $\vec{E}$ is also at $\vec{x}$ point. You never say $\vec{B}_{\vec{x}}$ but $\vec{E}_{\vec{y}}$. So this

reflects the principle of locality that the effect is propagating from point to point, you just have the derivative of the same point. You don't involve separate points. So we call Maxwell's equations as local equations, they contain only E and B at same point with finite number of derivatives. So the derivatives are the ones to help you to propagate because it relates the point to the neighboring point.

Another example of locality is the Einstein's general relativity. In Einstein gravity, the dynamical variables are so-called spacetime metric $g_{\mu\nu}(\vec{x}, t)$. They are object with two indices, two spacetime indices, and the Einstein's equations are equations for this kind of object $g_{\mu\nu}$. And again this equation are local equations, in the sense that they only depends on the g evaluated at the same point with a finite number of derivatives. So in Einstein's gravity, you no longer have action at distance, so the effect of the gravity is propagated through the spacetime.

Here we say these two examples exemplify the principle of locality. In fact, the principle of locality plays a very important role in formulating those equations, which is a very powerful principle.

Lecture 2: Classical Field Theories

Types of field

- A scalar field $\phi(\vec{x}, t)$. This is a quantity defined at each point; there is only one value at the given point. For example, the temperature is a single scalar quantity defined at each point.
- A vector field: $\vec{E}(t, \vec{x})$ and $\vec{B}(t, \vec{x})$ are examples of vector fields because you have a vector at each point. Here, I'm using the three-dimensional notations, but in the relativistic theory, we can use the four vector notations, e.x., the vector potential has the form $A_\mu(x^\nu)$. Here A_μ is a four-vector, obtained by combining the spatial and temporal components into a single spacetime vector.
- A tensor field: $g_{\mu\nu}(x^\lambda)$ is an example for tensor field. Here you have two indices, you have many different components depend on the dimension. At each point, you have an object with two indices.
- Spinor fields $\psi_\alpha(t, \vec{x})$, where α is another index that we will define later.

Convention

- μ is always from 0 to 3, and sometimes I write simply x ; when I write $x = x^\mu = (ct, \vec{x}) = (ct, x^i)$, it denotes a four-vector. The spatial index is always denoted by i .
- $\partial_\mu = (\frac{1}{c}\partial_t, \nabla)$, ∇ is the gradient on the spatial directions.
- A four-vector $A_\mu = (A_0, \vec{A}) = (A_0, A_i)$, we still use i to denote the spatial components.
- $c, \hbar = 1$.

Action Principle in Classical Mechanics

First, recall in your classical mechanics, we introduce the action

$$S = \int dt L(x, \dot{x}, t), \quad (2)$$

where Lagrangian is a function of x , $\dot{x}$ and the time. This is the one-dimensional particle motion. And you can also introduce the canonical momentum, which is defined by

$$p = \frac{\partial L}{\partial \dot{x}}. \quad (3)$$

The Hamiltonian is related to the Lagrangian by

$$H = p\dot{x} - L, \quad (4)$$

through a Legendre transformation. And the equation of motion is obtained by extremizing $S[x(t)]$, where S can be consider as a functional of your trajectory, so whatever trajectory you have, you extremize this S , and you obtain the equation of motion (EOM).

Classical Field Theory

Now we can generalize this to field theory. From the principle of locality, the form of the Lagrangian L in field theory is significantly constrained. It must have the form:

$$L = \int d^3x \mathcal{L}(\phi_a, \partial_t \phi_a, \partial_i \phi_a) \quad (5)$$

which is the integration over all spatial directions. Here, $\phi_a(\vec{x}, t)$ is a function of spatial direction and time, it is a general field, and a label different fields. e.g., a can label different scalar fields if you have different scalar fields, a can also refer to A_μ or to spinor fields, and so on. $\mathcal{L}$ is called the Lagrangian density, which depends only on the value of ϕ_a and its derivatives at a single point. They are still local.

Remarks: Why the Lagrangian must have this form?

- First, the principle of locality implies (5) only involves a single integral because the locality does not allow something like this,

$$L = \cdots + \int d^3x d^3y \phi(\vec{x}, t) \phi(\vec{y}, t) \quad (6)$$

which involving the ϕ at different points. If your Lagrangian contains such terms, then when we derive the equations of motion, they will not be local. EOM will involve the behavior of your field at one point and then influenced by the field value at a distant point, then you will not have local equation.

- We only allow first derivative in time, we don't allow high order derivatives in time. The reason is that again the equation of motion will involve more than two derivatives in time if your Lagrangian has high order derivatives in time. $\partial_t \phi$ in (5) implies that EOMs contain two t -derivatives, which come from our experiences. In real life, all experiments are determined by the initial conditions. For the initial condition, you only need to specify the position and velocity; no higher derivatives are required. If the equation of motion involves more than two derivatives in time, then you need to specify more general initial conditions. But for simplicity, for the most of the time, as we will see, we will restrict to quantum field theory in first derivative in time.

So now as in classical mechanics, we can introduce the canonical momentum, Hamiltonian, and so on. Here, we can introduce the so-called canonical momentum density

$$\Pi_a(\vec{x}, t) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a(\vec{x}, t)}, \quad \text{where} \quad \dot{\phi}_a(\vec{x}, t) = \partial_t \phi_a \quad (7)$$

So this is just the direct generalization of classical mechanics. The momentum density is evaluated at each point, so Π_a also depends on $\vec{x}$ and t . The reason we call this the momentum density is because here $\mathcal{L}$ is the Lagrangian density. In the same way, we can also define the Hamiltonian density associated with each degree of freedom (DOF):

$$\mathcal{H} = \Pi_a \dot{\phi}_a - \mathcal{L} \quad (8)$$

Then, the Hamiltonian you just integrate over all space

$$H = \int d^3x \mathcal{H} \quad (9)$$

essentially a sum over all degrees of freedom. So now we can talk about equation of motion. It is analogous to classical mechanics. The classical field theory is just like classical mechanics, you generalize it to infinite number of DOFs. So now with each point in space, you associate it with some DOFs. Again, you just do the variation of the action, you extremize the action to be 0,

$$\delta S = 0 \quad (10)$$

where the action is again defined by

$$S = \int dt L = \int d^4x \mathcal{L}(\phi_a, \partial_\mu \phi_a). \quad (11)$$

You can straightforwardly generalize this to include higher spatial derivatives. Now let us compute the variation:

$$0 = \delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \delta (\partial_\mu \phi_a) \right] = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \partial_\mu (\delta \phi_a) \right] \quad (12)$$

In this case, δ is the variation, and ∂_μ and δ are independent, so you can exchange them, i.e.,

$$\partial_\mu (\delta \phi_a) = \delta (\partial_\mu \phi_a) \quad (13)$$

Then, we can do the integration by part for the second term and obtain

$$0 = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) \delta \phi_a \right] + \text{boundary terms} \quad (14)$$

The boundary terms will be proportional to $\delta \phi_a$. We will always choose $\delta \phi_a$ such that boundary terms vanish. Equation (14) must vanish for arbitrary variations $\delta \phi_a(x^\mu)$, which can be any function of x^μ . Therefore, the prefactor has to vanish, i.e.,

$$0 = \frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right), \quad (15)$$

(15) is the general EOM for classical field theory.

Restrictions in Field Theory

From now on, we will make two restrictions for simplicity.

Translationally invariant

The first is that we restrict to field theory which are translationally invariant, which means the theory does not single out any special point in spacetime. If you perform the experiment here in Chengdu, it is the same as performing it in Washington DC. If the theory were not translationally invariant, experiments performed in Chengdu and Washington DC would have different results.

Lorentz invariant

The second is that we assume the field theory is Lorentz invariant. A Lorentz invariant quantity looks exactly the same for every observer moving at constant velocity, no matter how their coordinate axes are rotated or how fast they move relative to each other. A quantity is Lorentz invariant if it does not change under any Lorentz transformation that includes both: Spatial rotations (changing direction in space), and Lorentz boosts (changing velocity or reference frame).

Now let us look at some simple examples of classical field theories that satisfy these two conditions.

E.x. 1: Maxwell's theory.

The dynamical variable is the four-vector potential $A_\mu(x^\nu)$. For A_μ , you define the field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (16)$$

The electric and magnetic fields E and B can be obtained from $F_{\mu\nu}$. The action for E&M can be obtained by

$$S_{EM} = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) \quad (17)$$

This is the action for electromagnetism in the absence of charged matter. If you do the variation here to get the EOM, then you obtain the source-free (vacuum) Maxwell equations. This action is corresponding to a local theory due to a single integral of spacetime.